

Aim:

A. Create an Android application to design screens using different layouts and UI including Button, EditText, TextView, Radio Button etc.

B. Write an android application demonstrating response to event/user interaction for Checkbox, Radio button, Button, Spinner.

Creation of the project: New Project -> Empty View Activity -> Next -> Name of project/Application -> Location -> Language Kotlin -> Minimum SDK (API 24 Nougat Android 7.0) -> Build configuration language (Kotlin DSL) -> Finish

activity_main.xml creation: res -> right click -> New -> Android Resource Directory -> Select 2nd option -> layout -> OK

Layout Resource Directory: Right click layout -> New -> Sample Data Directory -> Layout Resource Directory

Gradle Script Configuration: Inside Gradle Scripts -> Open build.gradle.kts (Module: app) -> Add / verify activity XML

Main Activity File CODE:

```
package com.example.application1
import android.os.Bundle
import android.view.View
import android.widget.*
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContentView(R.layout.activity_main)
        // Linking UI components
        val nameInput = findViewById<EditText>(R.id.name)
        val submitButton = findViewById<Button>(R.id.button)
        val displayMessage = findViewById<TextView>(R.id.textViewResponse)
        val radioButtonGroup = findViewById<RadioGroup>(R.id.selectgender)
        val radioButton1 = findViewById<RadioButton>(R.id.radioButton1)
        val radioButton2 = findViewById<RadioButton>(R.id.radioButton2)
        val checkBox1 = findViewById<CheckBox>(R.id.checkBox1)
        val checkBox2 = findViewById<CheckBox>(R.id.checkBox2)
        val languageMessage = findViewById<TextView>(R.id.language)
        val spinner = findViewById<Spinner>(R.id.spinner)
        val classTextView = findViewById<TextView>(R.id.myclass)
        // Spinner data
        val year = arrayOf("FYCS", "SYCS", "TYCS")
        // Spinner adapter
        val adapter = ArrayAdapter(
            this,
            android.R.layout.simple_spinner_dropdown_item,
            year
        )
        spinner.adapter = adapter
        // Spinner selection listener
        spinner.onItemSelectedListener = object : AdapterView.OnItemSelectedListener {
            override fun onItemSelected(
                parent: AdapterView<*>?,
                view: View?,
                position: Int,
                id: Long
            ) {
                classTextView.text = "Your Class: ${year[position]}"
            }
            override fun onNothingSelected(parent: AdapterView<*>?) {
                classTextView.text = "Please select your Class"
            }
        }
    }
}
```

```

    }
}
// Gender selection
radioGroup.setOnCheckedChangeListener { _, checkedId ->
    when (checkedId) {
        R.id.radioButton1 ->
            displayMessage.text = "Gender: ${radioButton1.text}"
        R.id.radioButton2 ->
            displayMessage.text = "Gender: ${radioButton2.text}"
    }
}
// Submit button action
submitButton.setOnClickListener {
    val name = nameInput.text.toString()
    displayMessage.text = "Welcome, $name!"
    when {
        checkBox1.isChecked && checkBox2.isChecked ->
            languageMessage.text = "You like both Python and Kotlin"
        checkBox1.isChecked ->
            languageMessage.text = "You like Python"
        checkBox2.isChecked ->
            languageMessage.text = "You like Kotlin"
        else ->
            languageMessage.text = "No language selected"
    }
}
}
}
}

```

Activity_Main XML CODE:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">
    <EditText
        android:id="@+id/name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter your name" />
    <RadioGroup
        android:id="@+id/selectgender"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <RadioButton
            android:id="@+id/radioButton1"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Male" />
        <RadioButton
            android:id="@+id/radioButton2"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Female" />
    </RadioGroup>
    <CheckBox
        android:id="@+id/checkBox1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Python" />
    <CheckBox
        android:id="@+id/checkBox2"

```

```

        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Kotlin" />
<Spinner
    android:id="@+id/spinner"
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />
<TextView
    android:id="@+id/myclass"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Your Class will appear here" />
<Button
    android:id="@+id/button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Submit" />
<TextView
    android:id="@+id/textViewResponse"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Response will appear here"
    android:textSize="18sp" />
<TextView
    android:id="@+id/language"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Your language preference will appear here" />
</LinearLayout>

```

Build.gradle.kts(:app) CODE changes for the dependencies:

```

Dependencies{
    implementation("androidx.core:core-ktx:1.12.0")
    implementation("androidx.appcompat:appcompat:1.6.1")
    implementation("com.google.android.material:material:1.11.0")
    implementation("androidx.constraintlayout:constraintlayout:2.1.4")
}

```

OUTPUT:

